

Handling Process When a Bug Is Found or Regression Fails (QA Automation)

 Page is about?

Purpose

*This page outlines the **standard operating procedure** to follow when an automation **regression fails** or a **bug is detected** during scheduled test execution. The goal is to ensure consistent visibility, fast triage, and responsible CI/CD control to maintain quality and stability across environments.*

When to Trigger This Process

Trigger this process **immediately** when:

- A **critical test case fails** in daily regression that previously passed.
- A new or recurring **functional bug** is detected during automation.
- The failure **impacts release readiness**, reporting accuracy, or core workflows.

Standard Process Steps

1. Analyze & Confirm the Failure

- Verify whether the failure is **environmental**, **automation-related**, or a **functional defect**.
- Run the failing test **locally** to confirm repeatability, also make sure for given test data it works manually.
- Retrieve:
 - **Exception logs**
 - **Screenshots**
 - **Test Data**
 - **Execution report links**
 - **Relevant test case paths** (qTest/TestNg)

2. Log a JIRA Bug

- Create a detailed bug in the [QA Automation - Jira Board](#)
- Include:
 - Error message
 - Environment (Stage1, Stage 2, QC4)
 - Test case name/path
 - Screenshot and exception log links
 - Priority (Critical/High) and severity
 - GIT Change set, which might have caused this issue
 - Add appropriate labels/tags

3. Compose and Send Email Notification

Recipients:

- To: [AGS Tech Leads](#), [AGS Chapter Leads](#), [AGS Development](#), [Brian Danilczyk](#)
- Cc: Henry.Dittmer@ascensus.com, Phuong.Huynh@ascensus.com

Note: If [Brian Danilczyk](#) is out of office, [Valerie Gallegos](#) will coordinate follow-up actions.

Include in Email:

- Summary of failure
- Screenshots, logs, exception stack
- Links to:
 - JIRA ticket
 - Execution report (TestNG/qTest)
 - GitLab Merge Requests since last success
- **CI/CD Policy** for response (see below)



Email Template Draft (Generic)

Subject: Daily Regression Failed – [Module/Feature Name] | [Environment]

e.g. Daily Regression failed: NYD Enrollment Fails at Submit (Impacts Regression) | Stage 1

Message Body Draft:

Hi Team,

We encountered a critical issue during the latest automation regression run in [Environment] related to [Module/Feature].

The test failed due to an exception occurring at [Step Name].

Bug Summary:

- **Error on Screen:** "[Error Message]"
- **JIRA Bug:** [Link to Bug]
- **Regression Report:** [qTest/TestNg Execution Report Link]
- **Exception Log:** [Log File Link]
- **Screenshot:** [HTML Screenshot Link]
- **Test Data:** if applicable [Test Data TestNg Link]
- **Environment:** Stage1 / QC4
- **Priority:** High/Critical

Change Set Summary:

Below is a summary of recent **Merge Requests (MRs)** to the Monolith/QA Automation project during the timeframe between the **last successful run** and the **current failure**.

This helps identify potential root causes for regression failures:

```
===== Project: Monolith | Type: Merge Requests | Date Range: 07/30/2025 - 07/31/2025 | Total: 17 |  
Merged: 16, Closed: 1, Opened: 0, Other: 0 =====
```

Merged MRs During This Period:

- **MR Title:** Updated Enrollment Flow
- **MR Link:** [GitLab MR Link]
- **MR Author:** Author/Devs Name
- **MR Merged By:** Approver
- **Commit ID:** [Get Commit ID]
- **Status:** [Merged/Closed/Open]
- **Merged At (EST):** [Merge Date]

Note:

"**Author**" = Developer who initiated the MR

"**Merged By**" = Reviewer who approved and merged the MR

These entries are pulled from GitLab activity between the **last regression success timestamp** and the **current failure timestamp**.

MR Title: proxy change for librisk-web

MR Link: https://gitlab.com/ascensus-gs/products/depot/monolith/-/merge_requests/2634

Author: Chella Vignesh K P

Merged By: Priyal Patel

Commit ID: 5e861f14

Status: Merged

Merged At (EST): 07/31/2025 11:00 EDT

CI/CD Control Policy:

Main branch locked. Awaiting fix/rollback. Expect resolution or reversion **by 10:00 AM EST.**

Please inform us if this issue is already being tracked in JIRA, or if further context is needed to assist with resolution.

Thanks,

QA Automation Team

CI/CD Policy: Lockdown & Recovery

Lockdown:

- Immediately lock:
 - main branch of **monolith**
 - main branch of **automation**
- Swapnil Patil initiates lock

Resolution Deadline:

- Fix or rollback must be completed by **10:00 AM EST same business day**
- No delays. If fix is not ready, **revert** all changes made since the last successful run.

Manual Validation:

- Re-run only the failed test cases **locally**
- If passed, proceed with unlock
- Full suite regression is **not mandatory** for unlock if risk is low

Unlock & Notify:

- Unlock main branches after confirmation
- Send updated email with: details can be found [here](#)
 - Root Cause Analysis
 - Resolution steps taken (fix or rollback)
 - Retest confirmation

Additional Best Practices

Area	Best Practice
Bug Logging	Always link screenshots, logs, MR IDs, and test case path
Reporting	Always notify leads and maintain weekly report of blocked tests
Branch Lock	Enforce for all critical failures impacting release
Recovery	Validate manually; do not wait for Jenkins full run if local check passes